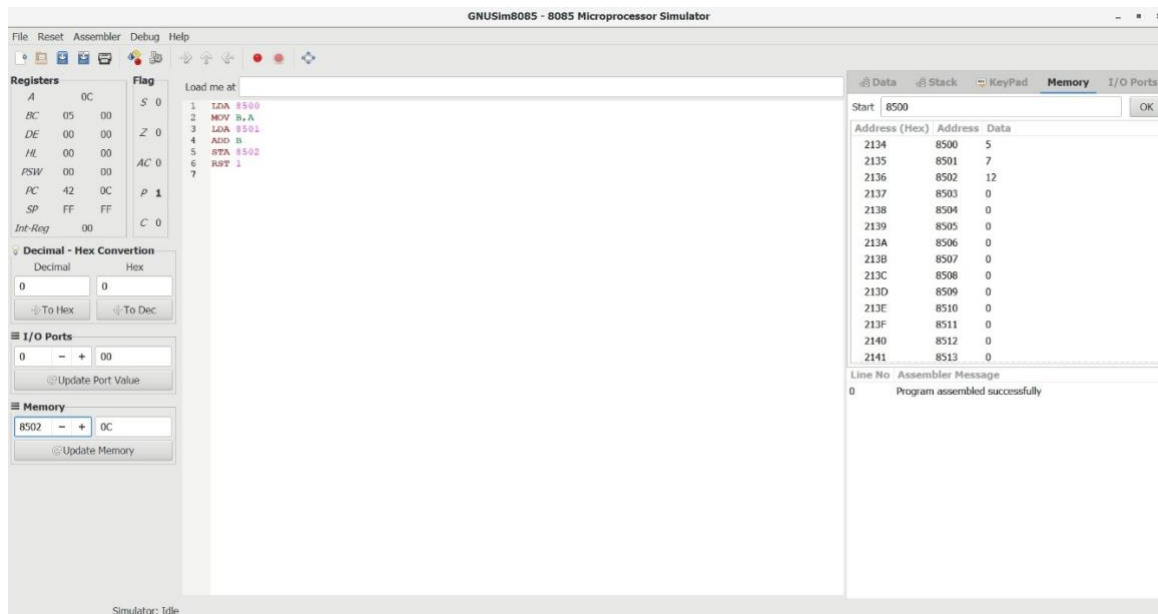


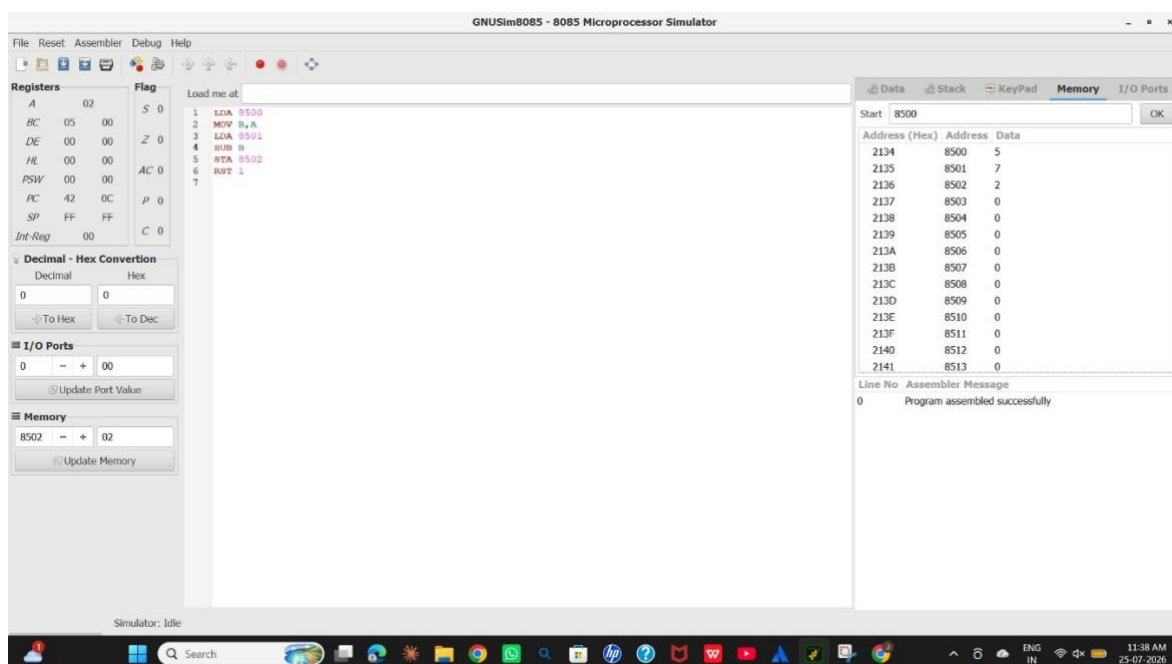
CSA1204 COMPUTER ARCHITECTURE

8085 LAB PROGRAMS

1. Addition of 2 numbers.



2. Subtraction of 2 numbers.



3. Addition with carry.

The screenshot shows the GNUSim8085 - 8085 Microprocessor Simulator interface. The assembly code in the center is as follows:

```
1 LDA 2500
2 MOV B,A
3 LDA 2501
4 STC
5 ADC B
6 STA 2502
7 HLT
```

The registers on the left show the following values:

Register	Value
A	00
BC	19 00
DE	00 00
HL	00 00
PSW	00 00
PC	42 0D
SP	FF FF
Int-Reg	00

The flags on the right show: S 0, Z 0, AC 0, P 0, C 0.

The memory window on the right shows the following data:

Address (Hex)	Address	Data
09C4	2500	25
09C5	2501	35
09C6	2502	61
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

The assembler message at the bottom right says: "Program assembled successfully".

4. Subtraction with borrow.

The screenshot shows the GNUSim8085 - 8085 Microprocessor Simulator interface. The assembly code in the center is as follows:

```
1 LDA 2500
2 MOV B,A
3 LDA 2501
4 SBC B
5 STA 2502
7 HLT
```

The registers on the left show the following values:

Register	Value
A	09
BC	19 00
DE	00 00
HL	00 00
PSW	00 00
PC	42 0D
SP	FF FF
Int-Reg	00

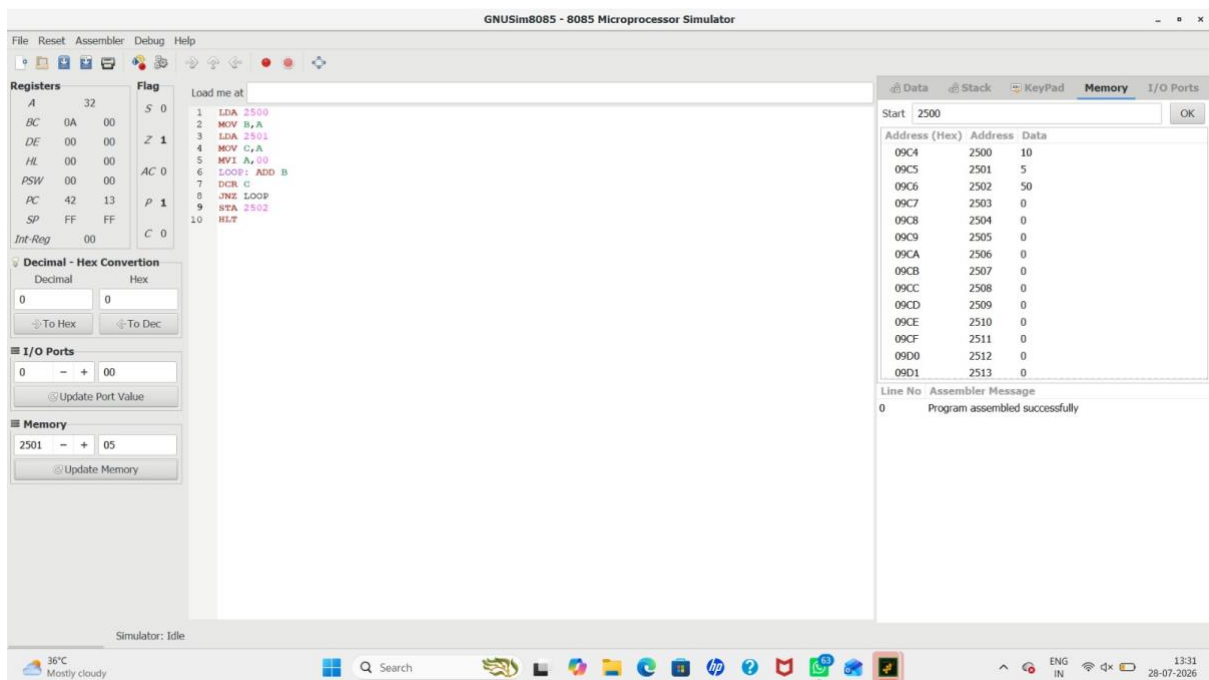
The flags on the right show: S 0, Z 0, AC 0, P 1, C 0.

The memory window on the right shows the following data:

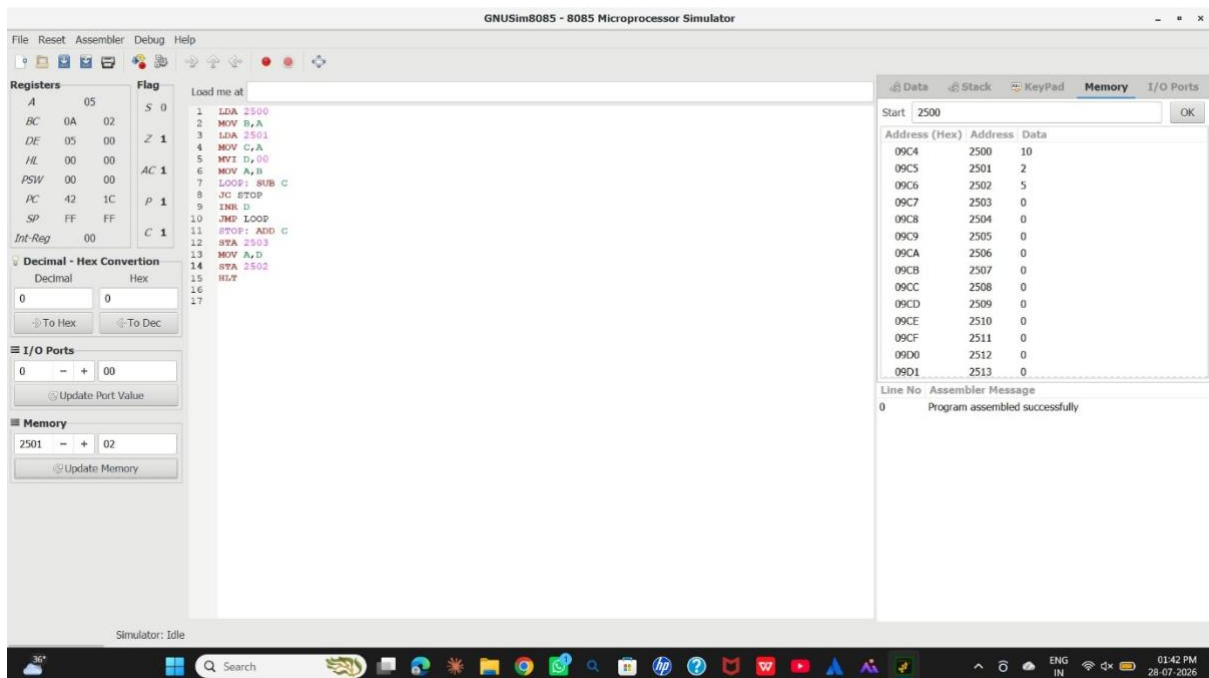
Address (Hex)	Address	Data
09C4	2500	25
09C5	2501	35
09C6	2502	9
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

The assembler message at the bottom right says: "Program assembled successfully".

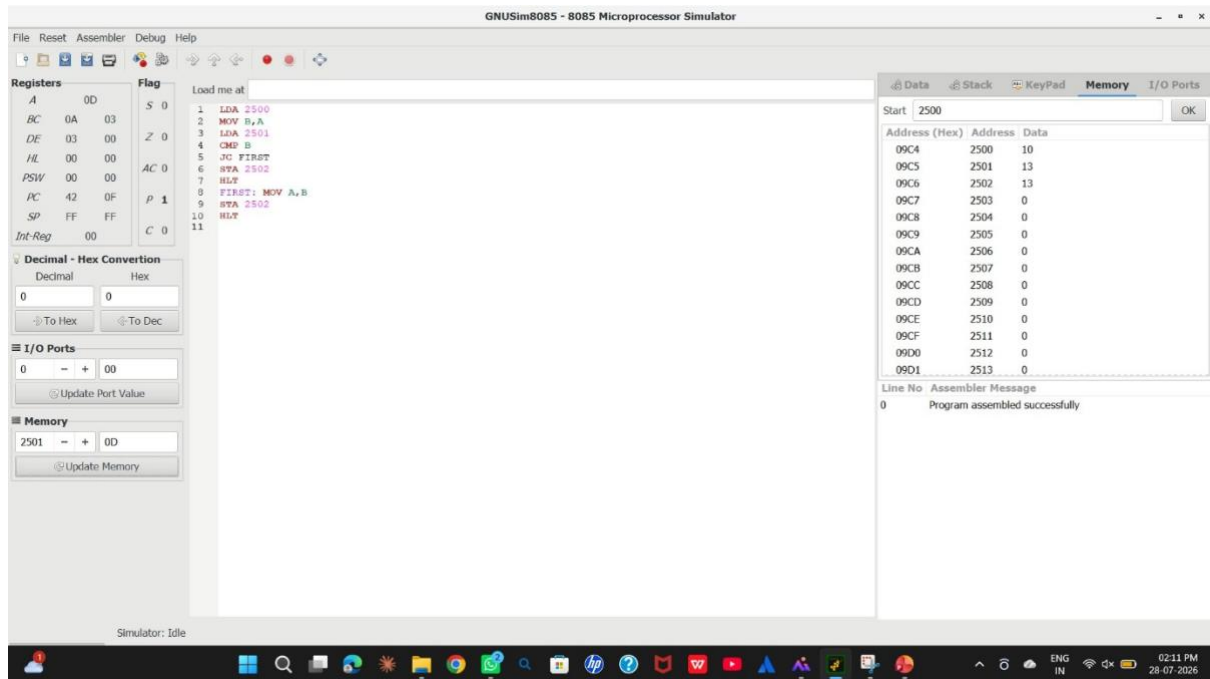
5. Multiplication of 2 numbers.



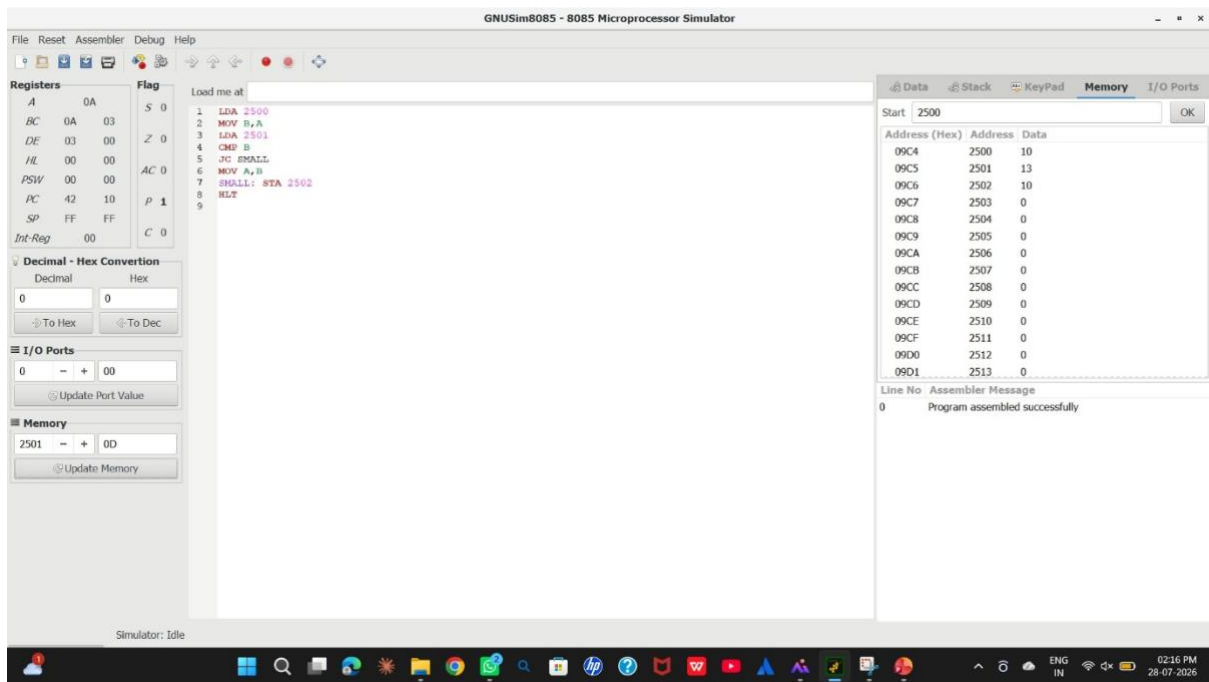
6. Division of two numbers



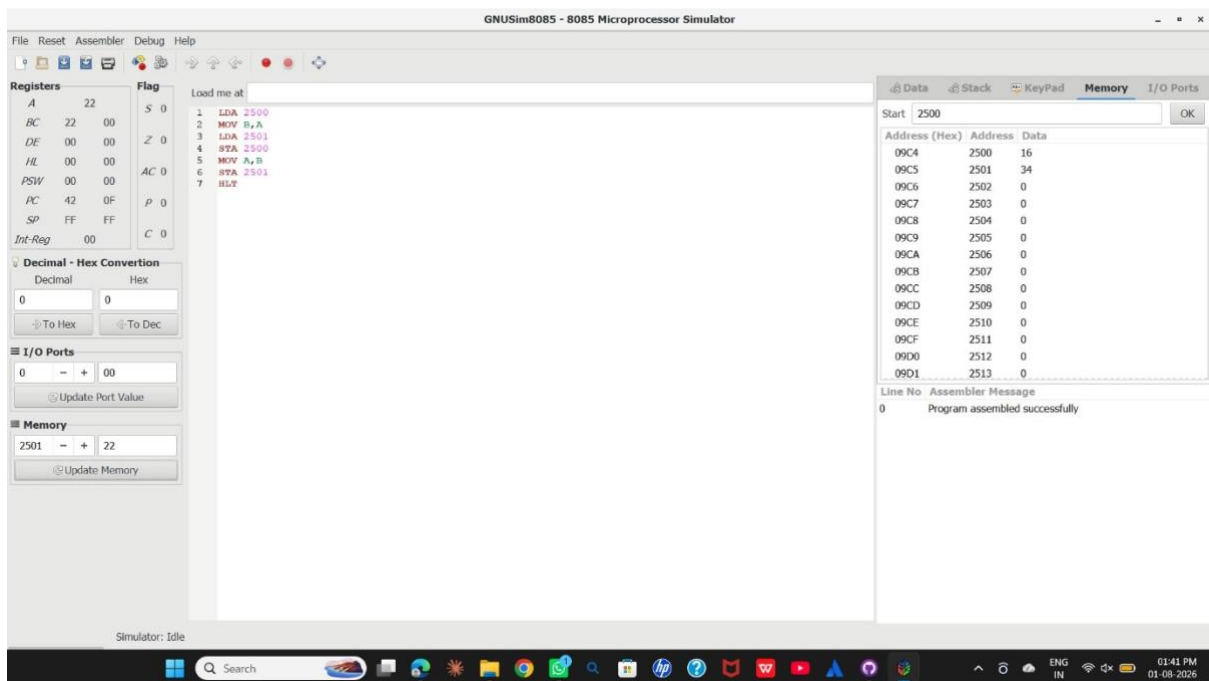
7. Greatest of 2 numbers.



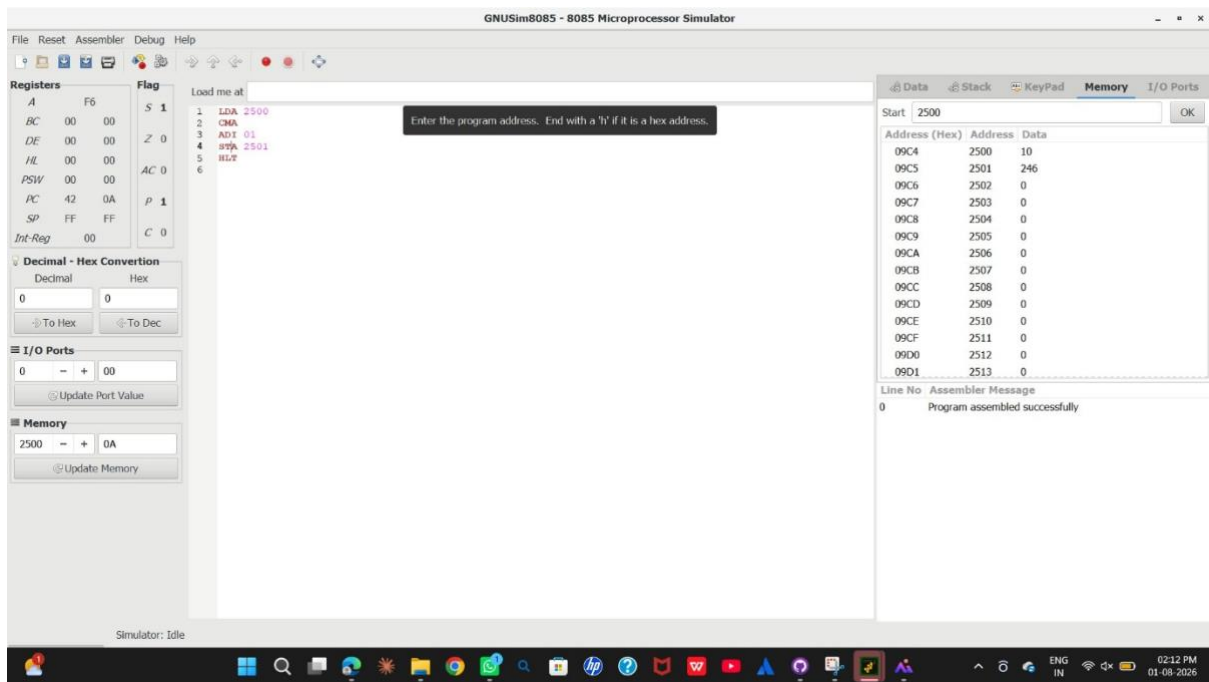
8. Smallest of 2 numbers.



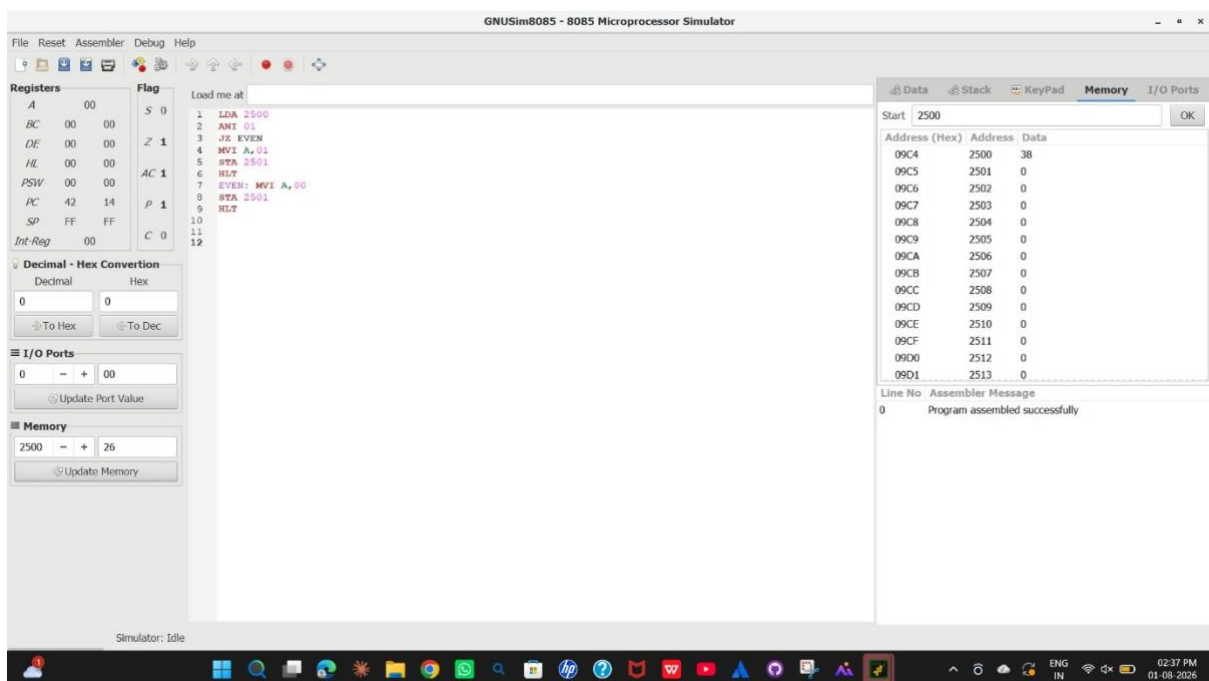
9. Swapping of 2 numbers.



10. 1's and 2's complement.



11. Given number is odd or even.



12. Given number is positive or negative.

